



Lab Mini Project On

Course Title : Microprocessor and Interfacing

Course code : CSE – 238

Submitted to

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Project 1: Student Result Calculator

Source code:

```
.MODEL SMALL
.STACK 100H

.DATA

    TITLEMSG DB 13,10,'STUDENT RESULT',13,10,'$'
    TOTALM DB 13,10,'TOTAL MARKS : $'
    AVGM DB 13,10,'AVERAGE : $'
    GRADEM DB 13,10,'GRADE : $'
    NEWLINE DB 13,10,'$'

    MATH DW 78
    PHY DW 82
    PROG DW 91

    TOTAL DW ?
    AVG DW ?

.CODE
MAIN PROC

    MOV AX,@DATA
    MOV DS,AX

    ; DISPLAY THE TITLE

    LEA DX,TITLEMSG
    MOV AH,09H
    INT 21H

    ; CALCULATE TOTAL MARKS

    MOV AX,MATH
    ADD AX,PHY
    ADD AX,PROG
    MOV TOTAL,AX

    PUSH AX

    ; DISPLAY TOTAL MARKS

    LEA DX,TOTALM
    MOV AH,09H
    INT 21H

    POP AX
    CALL DISPLAY

    ; CALCULATE AVERAGE MARKS
    MOV AX,TOTAL
    MOV BX,3
```

```

XOR DX,DX
DIV BX
MOV AVG,AX

PUSH AX

; DISPLAY AVERAGE MARKS
LEA DX,AVGM
MOV AH,09H
INT 21H

POP AX

CALL DISPLAY

; DISPLAY THE GRADE

LEA DX,GRADEM
MOV AH,09H
INT 21H

MOV AX,AVG

CMP AX,80
JAE GRADE_A

CMP AX,70
JAE GRADE_B

CMP AX,60
JAE GRADE_C

JMP GRADE_F

GRADE_A:
MOV DL,'A'
JMP PRINTGRADE

GRADE_B:
MOV DL,'B'
JMP PRINTGRADE

GRADE_C:
MOV DL,'C'
JMP PRINTGRADE

GRADE_F:
MOV DL,'F'

PRINTGRADE:
MOV AH,02H
INT 21H

MOV AH,4CH
INT 21H

MAIN ENDP

;DISPLAY NUMBER IN AX

DISPLAY:

```

```
PUSH AX
PUSH BX
PUSH CX
PUSH DX
```

```
MOV BX,10
XOR CX,CX
```

```
DIVIDE:
XOR DX,DX
DIV BX
PUSH DX
INC CX
```

```
CMP AX,0
JNE DIVIDE
```

```
PRINT:
POP DX
ADD DL,'0'
MOV AH,02H
INT 21H
```

```
DEC CX
CMP CX,0
JNE PRINT
```

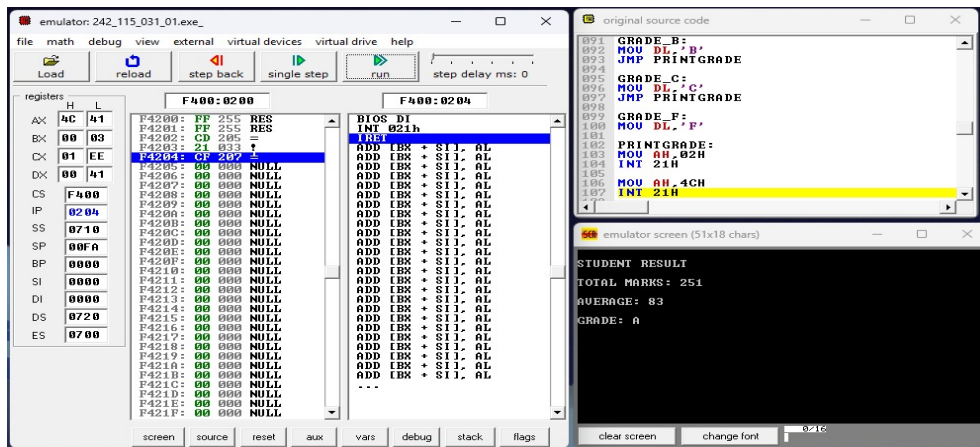
```
LEA DX,NEWLINE
MOV AH,09H
INT 21H
```

```
POP DX
POP CX
POP BX
POP AX
```

```
RET
```

```
END MAIN
```

OUTPUT:



Project 2: Menu-Driven Calculator

Source code:

```
.MODEL SMALL
.STACK 100H

.DATA
MENU DB 13,10,'SIMPLE CALCULATOR',13,10
      DB '1. Addition',13,10
      DB '2. Subtraction',13,10
      DB '3. Multiplication',13,10
      DB '4. Exit',13,10
      DB 'Enter Choice: $'
RESULT DB 13,10,'Result: $'
MSG DB 13,10,'$'

NUM1 DW 60
NUM2 DW 45

.CODE
MAIN PROC

      MOV AX,@DATA
      MOV DS,AX

START:
      MOV AH,09H
      LEA DX,MENU
      INT 21H

      MOV AH,01H
      INT 21H

      CMP AL,'1'
      JE ADDITION

      CMP AL,'2'
      JE SUBTRACTION

      CMP AL,'3'
      JE MULTIPLICATION

      CMP AL,'4'
      JE EXIT

      JMP START

;ADDITION OPERATION
```

ADDITION:

```
MOV AX,NUM1
ADD AX,NUM2
PUSH AX
```

;DISPLAY RESULT

```
LEA DX,RESULT
MOV AH,09H
INT 21H
```

```
POP AX
CALL DISPLAY
```

JMP START

;SUBTRACTION OPERATION

SUBTRACTION:

```
MOV AX,NUM1
SUB AX,NUM2
PUSH AX
```

;DISPLAY RESULT

```
LEA DX,RESULT
MOV AH,09H
INT 21H
```

```
POP AX
CALL DISPLAY
```

JMP START

;MULTIPLICATION OPERAION

MULTIPLICATION:

```
MOV AX,NUM1
MOV BX,NUM2
MUL BX ; RESULT IN AX
```

PUSH AX

;DISPLAY RESULT

```
LEA DX,RESULT
MOV AH,09H
INT 21H
```

```
POP AX
CALL DISPLAY
```

JMP START

DISPLAY:

```
PUSH AX
PUSH BX
```

```

PUSH CX
PUSH DX

;DIVISOR = 10

MOV BX,10

;CLEAR DIGIT COUNTER

XOR CX,CX

DIVIDE:

;CLEAR DX BEFORE DIVISION

XOR DX,DX

;DIVIDE AX BY 10

DIV BX

PUSH DX ;SAVE REMINDER

INC CX ;COUNT DIGIT

CMP AX,0 ; CONTINUE UNTIL QUOTIENT BECOME 0
JNE DIVIDE

PRINT:

POP DX
ADD DL,'0' ;CONVERT DIGIT TO ASCII

;PRINT DIGIT

MOV AH,02H
INT 21H

DEC CX
CMP CX,0
JNE PRINT

LEA DX,MSG
MOV AH,09H
INT 21H

POP DX
POP CX
POP BX
POP AX

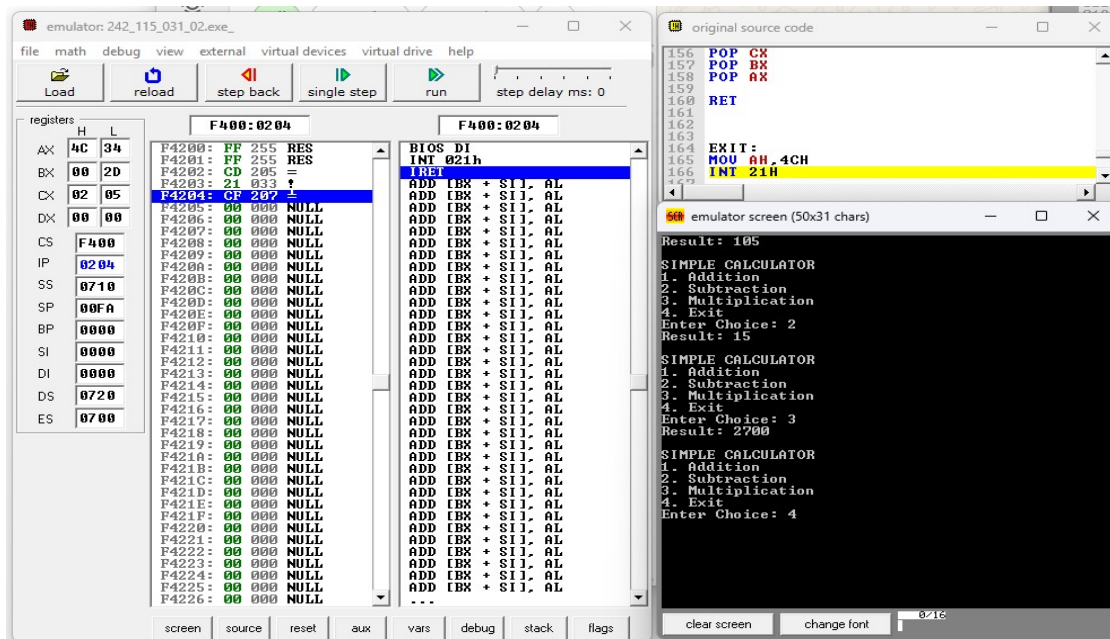
RET

EXIT:
MOV AH,4CH
INT 21H

MAIN ENDP
END MAIN

```

OUTPUT:



Project 3: Shopping/Billing System

Source code:

```
.MODEL SMALL
.STACK 100H
```

```
.DATA
```

```
MENU DB 13,10,'SHOP MENU',13,10
      DB '1. APPLE',13,10
      DB '2. ORANGE',13,10
      DB '3. MANGO',13,10
      DB '4. CHECKOUT',13,10
      DB '5. EXIT',13,10
      DB 'ENTER CHOICE: $'
```

```
APPLE_MSG DB 13,10,'APPLE ADDED$'
ORANGE_MSG DB 13,10,'ORANGE ADDED$'
MANGO_MSG DB 13,10,'MANGO ADDED$'
```

```
APPLE_Q DB 13,10,'APPLE QUANTITY: $'
ORANGE_Q DB 13,10,'ORANGE QUANTITY: $'
MANGO_Q DB 13,10,'MANGO QUANTITY: $'
```

TOTAL_MSG DB 13,10,'TOTAL BILL: \$'

APPLE DB 0
ORANGE DB 0
MANGO DB 0

TOTAL DW 0

.CODE

MAIN PROC

MOV AX,@DATA
MOV DS,AX

START:

LEA DX,MENU
MOV AH,09H
INT 21H

; TAKE CHOICE

MOV AH,01H
INT 21H

CMP AL,'1'
JE ADD_APPLE

CMP AL,'2'
JE ADD_ORANGE

CMP AL,'3'
JE ADD_MANGO

CMP AL,'4'
JE CHECKOUT

CMP AL,'5'
JE EXIT

JMP START

; ADD APPLE

ADD_APPLE:

INC APPLE

MOV AX,TOTAL
ADD AX,40
MOV TOTAL,AX

LEA DX,APPLE_MSG
MOV AH,09H
INT 21H

JMP START

; ADD ORANGE

ADD_ORANGE:

INC ORANGE

MOV AX,TOTAL

ADD AX,30

MOV TOTAL,AX

LEA DX,ORANGE_MSG

MOV AH,09H

INT 21H

JMP START

; ADD MANGO

ADD_MANGO:

INC MANGO

MOV AX,TOTAL

ADD AX,60

MOV TOTAL,AX

LEA DX,MANGO_MSG

MOV AH,09H

INT 21H

JMP START

; CHECKOUT

CHECKOUT:

; DISPLAY APPLE QUANTITY

LEA DX,APPLE_Q

MOV AH,09H

INT 21H

MOV DL,APPLE

ADD DL,'0'

MOV AH,02H

INT 21H

; DISPLAY ORANGE QUANTITY

LEA DX,ORANGE_Q

MOV AH,09H

INT 21H

MOV DL,ORANGE
ADD DL,'0'
MOV AH,02H
INT 21H

; Display Mango Quantity

LEA DX,MANGO_Q
MOV AH,09H
INT 21H

MOV DL,MANGO
ADD DL,'0'
MOV AH,02H
INT 21H

; DISPLAY TOTAL BILL

LEA DX,TOTAL_MSG
MOV AH,09H
INT 21H

MOV AX,TOTAL
CALL DISPLAY_NUM

JMP START

; EXIT

EXIT:

MOV AH,4CH
INT 21H

MAIN ENDP

; PROCEDURE TO DISPLAY NUMBER

DISPLAY_NUM PROC

MOV BX,10
MOV CX,0

NEXT:

MOV DX,0
DIV BX

PUSH DX
INC CX

```

CMP AX,0
JNE NEXT

```

PRINT:

```

POP DX
ADD DL,'0'

```

```

MOV AH,02H
INT 21H

```

```

LOOP PRINT
RET

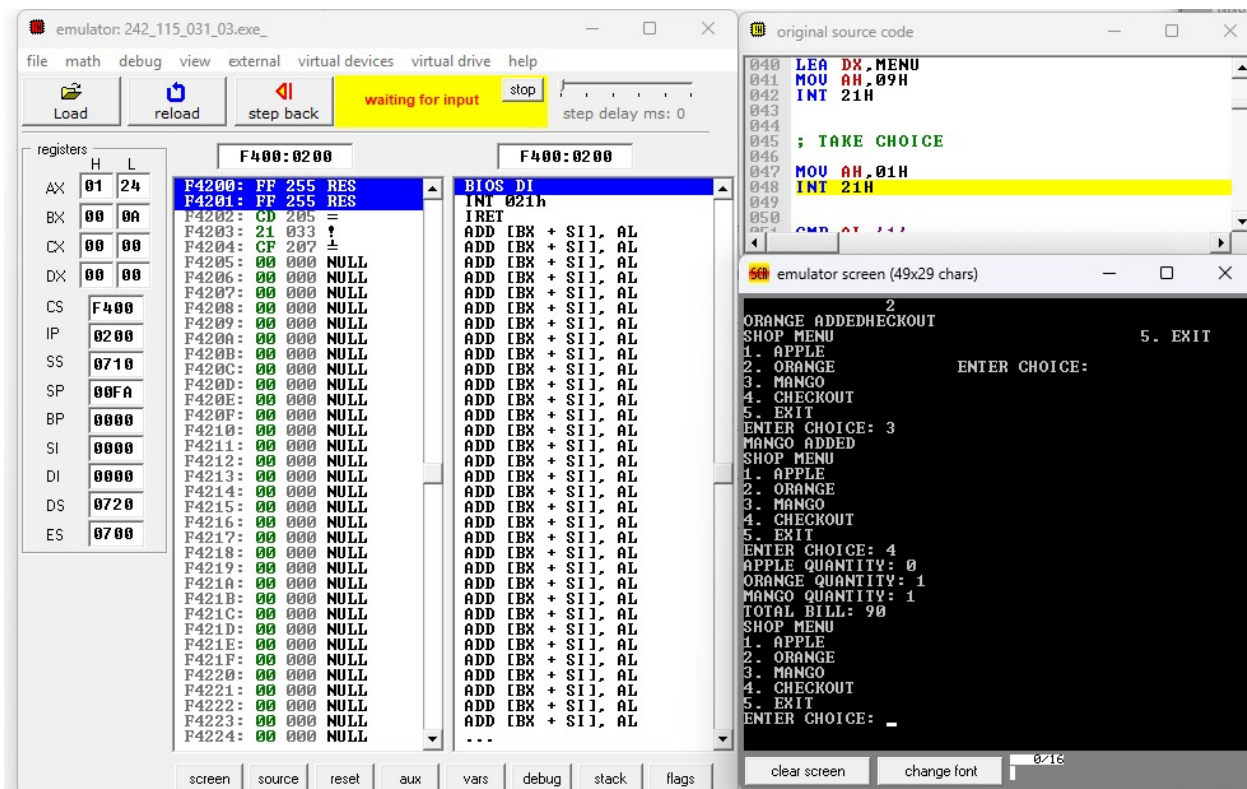
```

```

DISPLAY_NUM ENDP
END MAIN

```

OUTPUT:



Project 4: Password Verification System

Source code:

```
.MODEL SMALL
.STACK 100H

.DATA
    PASSWORD DB '5'      ; CORRECT PASSWORD
    ATTEMPTS DB 3         ; MAX ATTEMPTS

    MSG1 DB 13,10,'ENTER PASSWORD: $'
    MSG2 DB 13,10,'ACCESS GRANTED$'
    MSG3 DB 13,10,'WRONG PASSWORD$'
    MSG4 DB 13,10,'ATTEMPTS LEFT: $'
    MSG5 DB 13,10,'ACCESS DENIED$'

.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX

START:

    ; DISPLAY ENTER PASSWORD

    LEA DX,MSG1
    MOV AH,09H
    INT 21H

    MOV AH,01H
    INT 21H

    CMP AL,PASSWORD
    JE CORRECT

    ; DISPLAY WRONG PASSWORD

    LEA DX,MSG3
    MOV AH,09H
    INT 21H

    ;DECREASE ATTEMPT

    DEC ATTEMPTS

    ;CHECK IF ATTEMPTS == 0 THEN IT WILL SHOW ACCESS DENIED

    CMP ATTEMPTS,0
    JZ DENIED

    ; IT SHOWS ATTEMPTS LEFT
```

```

LEA DX,MSG4
MOV AH,09H
INT 21H

```

```

MOV DL,ATTEMPTS
ADD DL,'0'
MOV AH,02H
INT 21H

```

```

JMP START

```

CORRECT:

```

LEA DX,MSG2
MOV AH,09H
INT 21H
JMP EXIT

```

DENIED:

```

LEA DX,MSG5
MOV AH,09H
INT 21H

```

EXIT:

```

MOV AH,4CH
INT 21H

```

```

MAIN ENDP
END MAIN

```

OUTPUT:

